

.Program: Diploma in Civil and Environmental Engineering	
Course Code: 5352C	Course Title: Environmental Geotechnology
Semester: 5	Credits: 3
Course Category: Program Elective	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To provide the students with the basic concepts of contaminant transport mechanism in soil.
- To impart knowledge on soil – water- contaminant interaction and its effects on soil behaviour.
- To facilitate the students with the understanding about the design aspects of waste containment facilities and various remediation techniques for the contaminated site.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of Chemistry in Engineering	1004	Applied Chemistry	1
Knowledge of soil mechanics	4351	Soil Mechanics and Transportation Engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Understand Soil -water- contaminant interaction process and its importance in determination of soil behaviour.	10	Understanding
CO2	Know the basic concepts of contaminant transportation in soil and municipal solid waste management	10	Understanding
CO3	Understand the design aspects of waste containment facilities	12	Understanding

CO4	Plan remediation of contaminated sites	11	Understanding
	Series test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3			3		3		
CO4			3		3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand Soil -water- contaminant interaction process and its importance in determination of soil behaviour.		
M1.01	Explain the scope of geoenvironmental engineering	2	Understanding
M1.02	Understand the importance of geoenvironmental engineering in assessing the behaviour of soil in potential sites through case studies.	2	Understanding
M1.03	Understand the concept of doble layer	2	Understanding
M1.04	Describe the soil-water-contaminant interaction	2	Understanding
M1.05	Understand the change in properties of soil due to soil-water-contaminant interaction	2	Understanding

Contents:

Fundamentals of Geoenvironmental Engineering. Scope of geoenvironmental engineering - multiphase behavior of soil - applications – importance of soil physics, soil chemistry, hydrogeology, biological process – sources and type of ground contamination – impact of ground contamination on geoenvironment - case histories on geoenvironmental problems.

Soil-water-contaminant interaction and concepts of double layer –Soil mineralogy characterization– concept of double layer - importance of soil-water-contaminant interaction in determining soil behaviour - Change in properties of soil due to change in environment - Atterberg limits, shear strength, volume change, and permeability.			
CO2	Know the basic concepts of contaminant transportation in soil and municipal solid waste management		
M2.01	Explain the contaminant transport mechanism in soil	3	Understanding
M2.02	Explain the characteristics of municipal solid waste	2	Understanding
M2.03	Understand the Solid waste management rules	2	Understanding
M2.04	Understand the criteria for site selection for waste containment facilities	3	Understanding
Contents: Contaminant transport in soil -Transport process- Advection, Diffusion, Dispersion and sorption-Fick's equation. (basic concepts only) Characteristics of Municipal solid waste, Physical, Chemical and geotechnical characteristics-Identification of Hazardous and Non-Hazardous waste- waste dump and its impact on environment-Regulatory requirement -Solid waste management rules (brief introduction only) –MOEF&CC Guidelines-duties of waste generator and local authority – Evolution of waste containment facilities and disposal practices – Site selection based on environmental impact assessment–Hydrological consideration in landfill design			
CO3	Understand the design aspects of waste containment facilities		
M3.01	Identify the different types of landfills, landfill layout and capacity of landfill	2	Understanding
M3.02	Understand the layers of a landfill	2	Understanding
M3.03	Identify the components of landfill	2	Understanding
M3.04	Understand the criteria for material selection for the barrier layer	2	Understanding
M3.05	Explain the Gas removal and leachate collection systems in landfills.	2	Understanding
M3.06	Describe the closure and post closure monitoring systems in landfills.	2	Understanding
Contents:			

Waste Containment technology – Landfills - Landfill Types-Landfill layout and capacity Planning of landfills-Liner and Cover system, its components and its functions-natural clay liner- compacted clay liner selection of soil for barrier layer- Methods to find permeability of clay barrier layer -Primary and secondary leachate collection and removal systems - Gas Management, Gas extraction systems-passive and active system Closure and post closure monitoring system (brief introduction)			
CO4	Plan remediation of contaminated sites		
M4.01	Describe the application of geosynthetics in landfills	2	Understanding
M4.02	Understand the methodology of construction and design aspects of landfills	2	Understanding
M4.03	Risk assessment of contaminated site	2	Understanding
M4.04	Understand the soil exploration at contaminated site	2	Understanding
M4.05	Explain the in-situ and ex-situ remediation techniques	3	Understanding
Contents: Application of geosynthetics in landfills-Geotextile, geomembrane, geosynthetic clay liners, Geocomposites. methodology of construction, testing and design aspects Contaminated site- Soil exploration at contaminated site (brief introduction)-risk assessment of contaminated site - remediation methods for soil and groundwater –selection and planning of remediation methods. In-situ/ex-situ remediation - bioremediation, thermal remediation, pump and treat method, phyto remediation and electrokinetic remediation			

Text / Reference:

T/R	Book Title/Author
T1	Rowe R.K., "Geotechnical and Geoenvironmental Engineering Handbook" Kluwer Academic Publications, London, 2000.
T2	Reddi L.N. and Inyang, H. I., "Geoenvironmental Engineering, Principles and Applications" Marcel Dekker Inc. New York, 2000.
T3	Sharma H.D. and Reddy K.R., "Geoenvironmental Engineering: Site Remediation, Waste Containment, and Emerging Waste Management Technologies" John Wiley & Sons, Inc., USA, 2004.
R4	Daniel, D.E, "Geotechnical Practice for Waste Disposal", Chapman, and Hall,

	London, 1993
R5	Yong, R. N., "Geoenvironmental Engineering, Contaminated Soils, Pollutant Fate, and Mitigation" CRC Press, New York, 2001.
R6	Dr. G V Rao and Dr. R S Sasidhar, "Solid waste Management and Engineered Landfills", SaiMaster Geoenvironmental Services Pvt. Ltd. Publication, 2009
R7	Bagchi, A., "Design of landfills and integrated solid waste management" John Wiley & Sons, Inc., USA, 2004.
R8	Sparks, D.L., "Environmental Soil Chemistry" Academic Press, New York, 2002.
R9	Koerner, R.M., "Designing with Geosynthetics", Fifth Edition. Prentice Hall, New Jersey, 2005

Online resources:

Sl.No.	Website Link
1	https://archive.nptel.ac.in/courses/105/101/105101196/
2	https://archive.nptel.ac.in/courses/105/102/105102160/